Project

Homeopathy for preventing and treating health conditions: a protocol for an evidence evaluation

Prepared for

National Health and Medical Research Council

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Canberra ACT 2601

CONFIDENTIAL

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Dates

This Research Protocol received approval from the NHMRC Natural Therapies Working Committee (NTWC) on 13 July 2022.

History

The NHMRC has been engaged by the Department of Health (the Department) to update the evidence underpinning the *2015 Review of the Australian Government Rebate on Natural Therapies for Private Health Insurance* (2015 Review) (1). The natural therapies to be reviewed are Alexander technique, aromatherapy, Bowen therapy, Buteyko, Feldenkrais, homeopathy, iridology, kinesiology, naturopathy, Pilates, reflexology, Rolfing, shiatsu, tai chi, western herbal medicine and yoga. These therapies are amongst those excluded from the private health insurance rebate as of 1 April 2019.

To support the NHMRC in their evidence review, **HT**Analysts has been engaged to conduct a systematic review of the evidence of clinical effectiveness of homeopathy. Eligible studies received from the Department's public call for evidence, the Natural Therapies Review Expert Advisory Panel (NTREAP) and the NTWC will also be included in the evidence evaluation.

This Research Protocol has been developed by **HT**Analysts in conjunction with the NHMRC, the NTWC, and NTREAP to provide a framework outlining the methodology that will be used to review the evidence about homeopathy. It is intended that all associated materials will be developed in a robust and transparent manner in accordance with relevant best practice standards (2, 3).

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List of abbreviations

ARoH	Australian Register of Homoeopaths
ARTG	Australian Register of Therapeutic Goods
BRISA	Regional Base of Health Technology Assessment Reports of the Americas
CM	Complementary medicine
CI	Confidence interval
CINAHL	Cumulative Index to Nursing and Allied Health Literature
COMET	Core Outcome Measures in Effectiveness Trials
CPD	Continuing professional development
GMP	Good manufacturing practices
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HR	Hazard ratio
HTAnalysts	Health Technology Analysts
JBI	Joanna Briggs Institute
MD	Mean difference
MeSH	Medical Subject Headings
NHMRC	National Health and Medical Research Council
NRSI	Nonrandomised study of an intervention
NTREAP	Natural Therapies Review Expert Advisory Panel
NTWC	Natural Therapies Working Committee
OR	Odds ratios
PAHO	Pan American Health Organization
PICO	Population, Intervention, Comparator, Outcome
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RCT	Randomised controlled trial
RoB	Risk of bias
RR	Risk ratio
SD	Standard deviation
SMD	Standardised mean difference
SR	Systematic review
TGA	Therapeutic Goods Administration
TIDIER	Template for Intervention Description and Replication

1 Background

In 2015, a review of homeopathy found no reliable evidence demonstrating its efficacy in treating any clinical condition (4). The 2015 review was underpinned by an Overview of systematic reviews (SRs) that focussed solely on homeopathy and were published in the English language between 2007 and January 2013 (5, 6). Prospectively designed and controlled studies (i.e. randomised controlled trials [RCTs], quasi-randomised controlled trials, non-randomised controlled trials or prospective cohort studies) that were reported within included SRs and assessed homeopathy delivered to treat any clinical condition were included, with outcomes selected according to predefined criteria. The 2013 Overview informed the 2015 Review of the Australian Government Rebate on Private Health Insurance for Natural Therapies, which resulted in homeopathy and 15 other natural therapies being excluded from private health insurance rebates[1].

This review will not be limited by publication date and will primarily evaluate evidence reported in eligible RCTs. Evidence reported in NRSI, that meet minimum standards, will be included for populations where there is an absence of RCT evidence. The review will focus on evaluating the evidence for populations and conditions commonly seen and treated by homeopaths in Australia, including studies that assess homeopathy delivered for primary prevention.

The process for conducting the review is built upon the following framework:

1. source the clinical evidence by performing a systematic literature search,
2. identify the best available evidence published in English and indexed in English language databases,
3. incorporate additional literature identified through non-database sources received from the Department's public call for evidence, NTREAP and NTWC,
4. critically appraise and present the evidence, and
5. determine the certainty in the evidence base for each question, using a structured assessment of the body of evidence in accordance with GRADE methodology (7).

[1] *https://www.health.gov.au/resources/publications/private-health-insurance-reforms-changing-coverage-for-some-natural-therapies*

1.1 Description of condition and setting

Homeopathy has a long history in Australia, with the first practitioner having arrived in the 1840s followed by the establishment of the first of several homeopathic hospitals in 1876 (8). By the late 1920s, homeopathic hospitals ceased operation, primarily due to advances in medical practice such as the development of antibiotics. Nevertheless, over the last 50 years, homeopathy has experienced a revival, largely fuelled by changes in public perceptions regarding health (8, 9), with homeopathy typically used in Australia as an alternative to, or in conjunction with traditional medicine (10-15).

Studies exploring reasons why people in Australia use homeopathy are lacking, however individuals who typically seek out and use complementary medicines (CMs) are reported to be individuals who have a lower health-related quality of life due to chronic diseases or co-morbidities (12, 13, 15). CM product use in Australia is also reported to be higher among women, in full-time employment, and with a higher level of education (12, 13, 15). A 2002 study that assessed changes in the usage patterns and expenditure of alternative therapies in Australia found there was no difference in use of homeopathy between 1993 and 2000 (4.4% vs 4.3%) and no difference between the proportion of Australians who had visited a homeopath (1.2% for both surveys) (16, 17). More recent studies also suggest a continuing use of homeopathy in Australia (12, 13, 15).

Published reports indicate homeopathy is used to treat or alleviate symptoms associated with a range of chronic conditions such as allergic rhinitis, asthma, depression, rheumatoid arthritis, and irritable bowel syndrome (18-22), often when established treatments are not satisfactory, or when individuals are seeking to improve overall well-being (23, 24). Homeopathic medicinal products are also often used to prevent or treat side effects associated with cancer (25-27). In other cases, homeopathy is used by individuals seeking a more holistic lifestyle to treat infertility, or to prevent recurrent infections (28, 29). Given the breadth of the review, and variety of potential populations and conditions for which homeopathy is used (as found in the scoping review, see Figure 1), a concise description of each population and condition addressed will be provided after conduct of the full text review. See section 3.1.2 for information on the types of populations and conditions that are eligible for inclusion in the review and how they will be prioritised for data analysis and synthesis (if required).

Homeopathy can be practised in a range of settings (e.g., primary, acute, palliative care) (see section 1.2) and as such, this review will not be limited by setting.

Figure 1 Overview of potential populations/conditions (categorised by ICD-11 codes) for which homeopathy is used

Source: Homeopathy for preventing and treating health conditions: Scoping review

1.2 Description of intervention

Homeopathy is an alternative medical system that was first developed approximately 200 years ago by the German pharmacist Samuel Hahnemann (30). It is based on the premise "treat likes by likes", that is, if a substance causes similar symptoms in a healthy person that same substance can treat a disease with similar symptoms (31). The homeopathic system of treatment allocation and the recognition of clinical patterns of signs and symptoms differ from those of conventional medicine (32).

Patients seeking homeopathic treatment are likely to encounter various approaches depending on their homeopathy practitioner's philosophy and training (33). Homeopathy can be broadly categorised into two main types: individualised and non-individualised.

In individualised homeopathy patients are treated based on the totality of symptoms, with the most common form, known as 'Classical Homeopathy' involving the prescription of a single medicinal product , based on the holistic assessment of mental, emotional and/or physical symptoms (33). Following a detailed consultation, medicinal products are prescribed by a trained practitioner, matching a patient's symptoms with symptoms produced by these medicinal products in healthy individuals (34).

In non-individualised (or 'fixed') homeopathy, a specific homeopathic medicinal product is employed (with or without a consultation) for a specific condition, such that the same homeopathic medicinal product can be given to all patients with the same condition (35). For example all patients who have asthma would be given the same remedy (34). In addition, both single homeopathic medicinal products or 'complex' medicinal products can be prescribed, with complex medicinal products containing a fixed combination of multiple homeopathic ingredients (34). The most common forms of non-individualised homeopathy include 'Clinical homeopathy' and 'Isopathy'. In clinical homeopathy, complex medicinal products or single homeopathic remedies are prescribed based on the conventional medical diagnosis of the patient (36). In isopathy, allergens or causative infectious or toxic agents, including the patient's own bodily secretions (e.g. sputum or urine), are used to prepare the homeopathic medicinal product (34, 37).

Homeopathic medicinal products are prepared according to homeopathic pharmacopoeia and are created from a wide variety of substances such as plants, animals, minerals, or chemicals (32). To reduce toxicity and increase the effectiveness of homeopathic medicinal products, substances are serially diluted and vigorously shaken or agitated between each dilution (33). This process is called 'potentisation' (32).

Homeopathic medicinal products can be prepared as a liquid, powder (e.g. teething powder), granules or tablets (e.g. sugar pellets) (38) and are administered orally or applied externally (39).

Overview of the regulation of listed medicines and registered complementary medicines outlines the regulation of listed medicines and registered complementary medicines in Australia, including homeopathic preparations (40).

The *Therapeutic Goods Regulations 1990*[2] include the following definition for 'homoeopathic preparations':

homoeopathic preparation means a preparation:

[2] Available from: https://www.legislation.gov.au/Details/F2021C00839

(a) formulated for use on the principle that it is capable of producing symptoms in a healthy person similar to those which it is administered to alleviate; and

(b) prepared according to the practices of homoeopathic pharmacy using the methods of:

  (i) serial dilution and succussion of a mother tincture in water, ethanol, aqueous ethanol or glycerol; or

  (ii) serial trituration in lactose.

Most homeopathic medicinal products that are sufficiently dilute do not require listing in the Australian Register of Therapeutic Goods (provided they meet the requirements for exemption in Item 8 of Schedule 5 of the Therapeutic Goods Regulations 1990). However, they must still comply with other requirements under the Therapeutic Goods Act 1989 and the Therapeutic Goods Regulations 1990 (40). A homeopathic medicinal product prepared for a particular patient following a consultation (extemporaneous compounding) or to fill a prescription for that patient (dispensing) is also exempt from the requirement for ARTG listing.

Homeopaths are designated unregistered healthcare practitioners in Australia and are not required to be registered under legislation; however, homeopaths have established their own registration body which sets educational standards according to Government guidelines (36). The National Competency Standards in Homœopathy were established in 1999 to ensure practitioners of homeopathy are appropriately qualified and work within appropriate standards of ethical and professional behaviour to safeguard customers (41). The Australian Register of Homoeopaths (ARoH), also established in 1999, is the national registration board and self-regulation body for homeopaths (42). Certain criteria must be met to retain registration as a professional homeopathic practitioner with the ARoH including maintaining professional registration annually, meeting continuing professional development (CPD) requirements, and maintaining indemnity insurance (42). Prior to 2019, health fund rebates for homeopathic services were provided only to those who sought advice from practitioners who were registered with ARoH or members of certain associations (43).

Regulation of unregistered health practitioners (including homeopaths) varies across jurisdictions in Australia (44). In 2014, the Australian Health Ministers' Advisory Council published the National Code of Conduct for health care workers, a set of recommended provisions for health care complaints and enforcement powers (44). To date, New South Wales, Queensland, South Australia and Victoria have enacted legislation to implement the National Code (45). These regulatory regimes have been described as 'negative licensing', as they do not restrict who can provide health services (e.g. based on minimum qualifications) but enable disciplinary action to be taken against practitioners that fail to comply with minimum standards (45). Disciplinary action can include imposing fines and issuing orders that prohibit or set conditions on practice. Some states or territories that have not yet enacted the National Code, have provisions in existing legislation enabling complaints to be made about various unregistered health practitioners including homeopaths (for example, Tasmania) (44).

In Australia homeopaths are mostly self-employed in private, clinical practice in major towns and cities (41). As primary health care providers, homeopaths offer a unique, specialised and holistic approach to health and preventative healthcare (41). Members of the public may choose to consult with homeopaths in conjunction with conventional healthcare providers to broaden the range of available treatment options. In addition, to prescribing homeopathic medicinal products ,

homeopaths also integrate several other techniques to facilitate the homeopathic therapeutic approach, such as nutritional guidance, personal hygiene, and counselling (41). Generally, homeopaths require long consultations, usually at least one hour, whereby all aspects of a patient's illness and life are considered, and then an appropriate treatment is chosen based on the patient as a whole and not on the illness or symptoms alone (46). Homeopathic treatment is also employed as a therapy of trade by a range of non-homeopathic practitioners, for example naturopaths (47).

1.3 How the intervention might work

Homeopathy is based on the core tenet of similitude ("treat likes by likes" or "like cures like"). The principle of 'similars' is based on the idea that substances which can elicit specific symptoms in healthy individuals, can be used to treat patients and/or diseases with similar symptoms. Homeopathic medicinal products are created by a series of dilutions (usually 1:10 or 1:100 diluent: volume ratio) with agitation between each dilution, called "succussions" (48). Succussion is believed to render highly diluted solutions biologically active (49). Serial dilutions may occur more than 30 times before the final homeopathic medicinal product is produced. Established potency scales can be the centesimal or "C scale", the decimal or "D scale", or the quintamillesimal or "Q/LM scales". The serial kinetic dilution is called "potentisation". Both high potency (higher dilution) and low potency (lower dilution) homeopathic medicinal products are commonly used (50).

The scientific basis for homeopathy is subject to substantial debate, and the mechanism of action of homeopathic medicinal products has not been established. In some cases substances are diluted beyond the limit of Avogadro's number, such that it is unlikely even one molecule of the original substance is present (33). Various explanations have been proposed for how highly diluted solutions may be biologically active after succussion (51-53). For example, water clusters or clathrates may form around the original molecules and remain even after those molecules are removed (54). These are then hypothesised to be recognised within the body to elicit a response. Alternatively, it has been suggested that nanoparticles of the original material may remain and cause a response (55).

Individualised prescribing introduces unique requirements when designing and conducting clinical research in homeopathy, given that patients with the same diagnosis are treated with different homeopathic medicinal products based on their characteristic symptoms.

1.4 Why it is important to do this review

In Australia, natural therapies, including homeopathy, are most often used in conjunction with conventional medicine and other strategies for supporting good health and wellness. Homeopathy is a popular CM in Australia, with a 2005 survey reporting 6% of respondents had used homeopathy and, of those 6%, 47.7% had visited a homeopathic practitioner in the last 12 months (15). For these reasons, it is important to synthesise the evidence for the effectiveness of homeopathy, to enable consumers, healthcare providers and policy makers to make informed decisions about care.

The 2013 Overview of systematic reviews (sometimes called an Umbrella review) identified 57 systematic reviews published in the English language between 2007 and January 2013. The Overview concluded that there was no reliable evidence that homeopathy is an effective treatment for any of the reported clinical conditions (5, 6). Overviews or Umbrella reviews are an efficient way to collect and evaluate evidence. However, they are dependent on the quality and completeness of the included systematic reviews and may be less comprehensive than systematic reviews of primary literature.

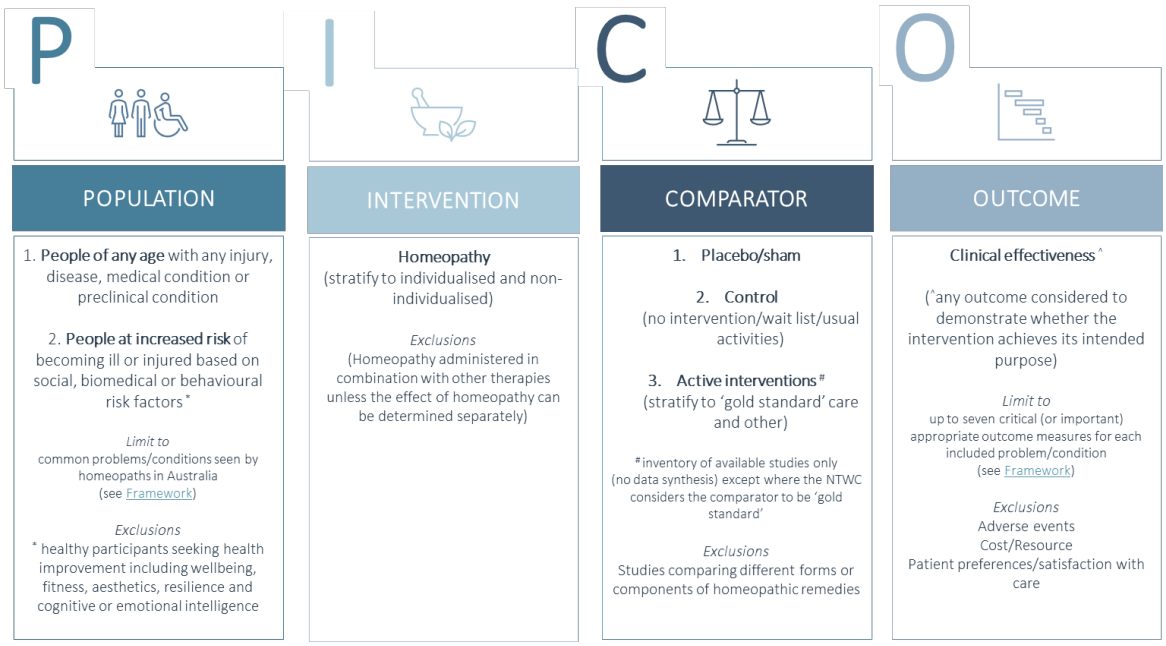
This systematic review will evaluate primary studies (e.g. RCTs) relating to the effectiveness of homeopathy for conditions commonly seen and treated by homeopaths in Australia. There will be no limit based on publication date. The rationale for conducting this review is to inform the Australian Government's Natural Therapies Review, which is evaluating evidence of the clinical effectiveness of 16 natural therapies (including homeopathy).

2 Objectives

The overall objective of this systematic review is to evaluate the effectiveness of homeopathy in preventing and/or treating injury, disease, medical conditions, or pre-clinical conditions. As illustrated in Figure 2, this review will focus on evaluating the evidence for populations and outcomes commonly seen and treated by homeopaths in Australia, and relevant to inform the Australian Government Natural Therapies Review for private health insurance rebates. It is out of scope of this review to evaluate the safety or cost-effectiveness of homeopathy.

Primary and secondary objectives of this review are outlined in the PICO framework below (see Figure 2). Specific research questions will be refined through a staged prioritisation process (outlined at Appendix A – Frameworks) that align with the research objective and are consistent with the approach used across the evidence evaluations of other natural therapies examined in the Review of Natural Therapies.

Figure 2 PICO framework for the research objective



3 Methods

Methods reported in this protocol are based on that described in the *Cochrane Handbook for Systematic Reviews of Interventions* (56) and relevant sections in the Joanna Briggs Institute (JBI) Reviewer's manual (57). Covidence (www.covidence.org), a web-based platform for producing systematic reviews, will be used for screening citations and recording decisions made. Covidence is compatible with EndNote and Microsoft Excel, which will be used for managing citations and data extraction, respectively. Where appropriate, RevMan (58) will be used for the main analyses. GRADEpro GDT software (www.gradepro.org) will be used to record decisions and derive an overall assessment of the certainty of evidence for each outcome guided by GRADE methodology (7). The final approved review protocol will be registered on the international prospective register of systematic reviews (PROSPERO).

To identify the evidence base for the clinical question, a systematic search of published literature will be conducted. All potentially relevant RCTs will be identified after applying pre-specified inclusion and exclusion criteria as outlined below. Independently from the evidence reviewers, the NTWC and NTREAP will determine a list of up to 20 populations considered a priority for the review. For populations where no RCT evidence has been identified (see Sections 3.1.1, 3.1.2 and 3.1.4), evidence published in NRSIs will be sought and assessed (where eligible and available) (See Section 3.2).

3.1 Criteria for considering studies for this review

3.1.1 Types of studies

Study design

Eligible studies are primarily RCTs that examine the effectiveness of homeopathy compared to control or another intervention. NRSIs are eligible for inclusion in instances where no RCT/s are identified for a prioritised population.

Randomised controlled trials

The primary study of interest is an RCT. Cluster-randomised trials and crossover trials are also eligible for inclusion, and will be analysed as described in Section 3.3.9 (59). If the method of randomisation is not specifically stated, or not strictly random (e.g., alternate allocation), then the study will be judged to be quasi-randomised. Quasi-RCTs will be evaluated alongside RCTs, with methods of randomisation examined in the risk of bias assessment (see Section 3.3.7) and any concerns about risk of bias addressed in the synthesis.

Nonrandomised interventional studies

NRSIs will only be eligible for inclusion for a priority population where no RCTs are available. In this case the NRSIs must also include design features as described below to be eligible.

To be eligible, the minimum design features of an NRSI include the following:

- allocation to, or practice of, the intervention occurs by choice (by the participant or other)
- the effect of the intervention in individuals (or clusters of individuals or groups) is compared with a concurrent control group
- researchers used methods to control for confounding, either:
    - in principle (for any confounding)
    - in principle (for time invariant unobserved confounding), or
    - for confounding (by observed covariates)
- potential confounders were measured before the intervention

NRSIs in which the effect of the intervention is compared to a historical (or non-parallel or non-concurrent) control group are not eligible for inclusion due to concerns of bias (e.g., due to residual confounding or unmeasurable changes in clinical practice over time).

Single arm studies (e.g. case series with post-test or pre-test/post-test outcomes), cross-sectional studies and case reports are not eligible for inclusion, as the features of these study designs make it difficult (if not impossible) to attribute observed changes in outcomes at this level.

Eligible NRSIs that are assessed to be at serious or critical risk of bias for one or more domains (see Section 3.3.6) will not be included in the evidence synthesis because results from these studies are likely to lead to misinformed judgements about the effect estimate.

The decision to limit the inclusion of NRSIs to priority populations where there is an absence of RCTs, was informed by scanning results from a scoping search of the published literature indexed in Embase, PubMed and the Cochrane Central Register of Controlled Trials (CENTRAL). The scoping search used the search strategy described in Appendix B, which retrieved more than 5000 potentially eligible RCTs (after the removal of duplicates). We anticipate a good proportion of the RCTs (250-400) identified in the scoping search will be eligible for inclusion in the review. Therefore, it is unlikely that inclusion of NRSIs will increase the certainty of the results. However, inclusion of

NRSIs where no RCT evidence is found aims to ensure the breadth of the PICO across priority populations is assessed.

Publication date

There are no limitations on publication date, however studies published after the systematic review literature search date will not be eligible for inclusion. Studies that are published (or submitted to the Department) after the literature search date will be listed within the *'Studies awaiting classification'* table of the evaluation report. These studies will not be subject to a formal evidence evaluation, however, a brief statement about the study and its potential impact on the overall conclusions of the evidence review will be included under the relevant sections of the review (e.g. 'Overall completeness and applicability of evidence').

Studies published in languages other than English

The literature search, as well as the Department's call for evidence, will not be limited by language of publication. Non-English databases will not be searched, however studies in languages other than English may be identified via the English-language databases. For pragmatic reasons, potentially eligible studies will not undergo full-text translation or data extraction but will be documented via a process outlined in Section 3.3.1 *Studies published in languages other than English*. The implications of not translating these studies will be considered when assessing bias arising due to missing results in the GRADE assessment (see Sections 3.3.13 and 3.3.16).

3.1.2 Types of participants

People of any age with any injury, disease, medical condition or pre-clinical condition are eligible for inclusion. This includes disease prevention in 'at-risk' healthy populations, which is broadly defined as those who are at increased risk of becoming ill or injured based on social, biomedical or behavioural risk factors (64). For a study to be considered eligible as 'at risk,' the study must provide sufficient evidence[3] of the individual participant being 'at risk,' not a population in general (e.g., a study that enrols otherwise healthy student nurses and examines the effect of the intervention on stress or burnout would be excluded, but a study that enrol student nurses who meet an enrolment criterion of elevated perceived stress would be eligible). For the purposes of this review, social determinants include factors such as income, education, employment and social support; biomedical factors include a person's health status (such as obesity, high blood pressure, high cholesterol, age, vitamin deficiency) and genetic make-up; and behavioural factors include a person's lifestyle choices (e.g. alcohol consumption, diet, exercise, tobacco and other drug use, etc.).

Healthy participants seeking health improvement, such as general wellbeing, fitness, aesthetic improvements, resilience and cognitive or emotional intelligence are not eligible for inclusion; however, a study with eligible and ineligible populations will be included if separate data is available for the eligible population/s (e.g., a study that enrols apparently healthy students and reports the effect of the intervention on anxiety would be excluded, except where separate data are available for students who have elevated symptoms of anxiety at enrolment).

Populations and conditions that are identified as eligible for inclusion in the review will be grouped according to ICD-11 codes[4] in the first instance. This is to help facilitate management of the eligible

[3] defined as a threshold minimum requirement of an individual participant, not the broader population group.

[4] See *https://icd.who.int/browse11/l-m/en*

studies, aid in the understanding of the population and to help determine the most appropriate place a study would contribute for evidence synthesis (i.e. to ensure the same data was not used in the analysis across multiple conditions, and to minimise heterogeneity).

ICD-11 categories will be based on the primary clinical condition reported in the study, and will be assigned prior to any critical appraisal, data synthesis or review of study size or results. The ICD-11 categories are considered sufficiently flexible to enable this approach (e.g. ICD-11 Category 21: 'Symptoms, signs or clinical findings, not elsewhere classified' could encompass all chronic pain populations including low back pain, neck pain, chronic cancer related pain, and fibromyalgia).

To ensure the review is manageable, data analysis will be targeted to up to 20 priority populations identified by NTREAP and NTWC (using PRACI and Australian equivalent survey data) as most relevant to the practice of homeopathy in Australia. The proposed population and condition prioritisation process is outlined in Appendix A – Frameworks (Flowchart 2). Decisions about which groups to include in the final analysis will be made through the prioritisation process, which may lead to changes in the population groups (i.e. becoming broader or narrower, or introduction of new groups).

The evidence reviewers will screen literature for eligible RCTs and compile a list of all populations and conditions with RCT evidence available. Independently from the literature screening, NTWC and NTREAP will rank populations to derive up to 20 priority populations for review. Should any of the priority populations have no RCT evidence available, the evidence reviewers will screen literature for NRSIs in these priority populations only.

The types of populations and conditions so far identified in the scoping search are outlined in Figure 1. Populations not prioritised for analysis and synthesis will be listed in an evidence inventory in the technical report.

For data synthesis, grouping by population/ condition will be proposed for consideration and agreement by NTWC. Given the results of the scoping search, we anticipate that grouping by umbrella population (e.g., gastrointestinal complaints) and condition (e.g., IBS, IBD, infantile colic) will make the best use of the available data and be consistent with how homeopathy is prescribed and used by homeopaths in Australia. However, to ensure the best use of available data, confirmation of grouping for data analysis and synthesis will occur at population prioritisation in consultation with NTWC and prior to data extraction.

3.1.3 Types of interventions

Intervention

Any homeopathic treatment or homeopathic medicinal product is eligible for inclusion. This includes simple homeopathic medicinal products involving single substances, and complex medicinal products involving more than one substance.

The treatment may be individualised (i.e., prescribed by a homeopath according to the person's presenting symptoms after a consultation) or non-individualised (i.e., where the same homeopathic medicinal product is given to all patients with the same condition, with or without a consultation).

There are no limits on the type of homeopathic preparations (i.e., sublingual sugar pellets, ointments, gels, drops, creams, sprays, and tablets). However the homeopathic medicinal product

must be administered orally or applied externally (i.e. topical, oral, nasal, rectal, vaginal, ocular or auricular use) , with parenteral use (by injection) excluded (39).

To allow for potential subgroup analyses (and to inform decision-making), studies will be stratified based on whether the participants received individualised or non-individualised homeopathy (see 3.3.15).

Homeopathic products that contain other 'active' ingredients (e.g. nutritional, herbal or pharmaceutical) will be excluded, however, excipients/non-active ingredients are acceptable.

Comparators

There are no restrictions on the type of eligible comparators, noting that the review will stratify the evidence into three comparisons: (i) placebo/sham (ii) inactive control (including no intervention, wait list or usual activities [if considered inactive]); and (ii) other 'active' interventions (inclusive of usual care if considered active).

Where usual care is poorly described it will be considered an 'inactive' comparator (i.e., homeopathy versus usual care [inactive]). Where usual care is delivered as an adjunct to homeopathy (i.e., homeopathy plus usual care versus usual care alone), the study will also be considered alongside those studies that use an inactive intervention. Other 'active' comparators may include (but will not be limited to) pharmacologic treatments, manual therapies, exercise programs, or other forms of physical activity designed to improve health.

Evidence synthesis comparing homeopathy to either placebo/sham or inactive control will be the main comparison reported. We will consider the two comparisons as subgroups within the same meta-analysis where appropriate. The decision to separate placebo/sham from inactive control is considered necessary to differentiate any placebo effect, particularly in patients who may be highly motivated to receive homeopathic treatment.

Evidence synthesis of 'active' interventions will only occur in exceptional circumstances (see Section 3.3.11). For studies involving other active comparators, we will present an 'evidence inventory' to provide a snapshot of the available evidence comparing homeopathic medicinal products with other 'active' interventions. These studies will be presented in a table, with a brief description of the PICO for each study.

In addition to the studied treatment and control intervention, co-interventions (e.g. diet, education programs, life modification, or medication) may be administered simultaneously to the treatment and control group. Studies with co-interventions will be included if all arms of a study receive the same co-interventions (i.e., the effectiveness of homeopathy is not confounded).

Head-to-head studies comparing different types or forms of homeopathy (e.g., individualised versus non-individualised, tablets versus sublingual pellets), different dilutions, potency or dose of the same homeopathic medicinal product, or different homeopathic medicinal products (e.g., simple versus complex) will be excluded. This is because the main objective of the review is to examine the effects of homeopathy, rather than the comparative effects of different homeopathic medicinal products (37).

3.1.4 Types of outcome measures

Outcome role

Outcomes will not be used as a criterion for including or excluding studies.

Outcome domains of interest

Outcomes are intended to align with the reasons why patients use the therapy and/or practitioners prescribe the therapy. This includes management of signs or symptoms associated with a clinical condition (such as chronic pain associated with fibromyalgia, rheumatoid arthritis, dysmenorrhoea, gastrointestinal disorders, or depression associated with a mood disorder), reduction in the need for, or side effects associated with traditional therapies (e.g., antihistamine use in allergic rhinitis, nausea and fatigue in cancer), recovery from, or changes in, disease outcomes (e.g., improved lung function in asthma, reduction in the number of upper respiratory tract infections in COPD), improvement in psychological/behavioural symptoms (e.g., depression, anxiety, stress) or overall health related quality of life.

It is out of scope of this review to assess personal health care preferences, patient experience measures (PREMS) (e.g., satisfaction with care), safety, quality and economic outcomes.

Outcome measures and timepoints of interest

Any effectiveness outcome anticipated to demonstrate a treatment achieves its intended purpose is eligible for inclusion (61, 65). There are no limitations on time points (e.g., short and long-term outcomes) or outcome measure. That is, all objective and subjective measures are eligible; including clinical and laboratory assessments (preferably accepted surrogate outcome measures such as HbA1C for diabetes, body mass index for weight gain or loss, lung function tests for asthma) and patient-reported outcome measures (PROMS) (preferably measured using validated tools).

As there are a broad range of populations eligible for inclusion in the review, it is not possible to pre-specify outcomes in this protocol, but these will be developed in a blinded process as described below.

All outcomes measured (or pre-specified in protocols or clinical trial registries) in each eligible RCT (or NRSI) will be listed in the '*Characteristics of included studies*' tables. Results will be extracted for those outcomes identified as critical or important to this review. For each priority population, results for a maximum of seven critical (or important) outcomes will be meta-analysed and reported in GRADE summary of findings (SoF) tables with corresponding evidence statements (see Section 3.3.17).

Identification of the critical (or important) outcomes will occur after identification of eligible priority populations/conditions. To avoid introducing bias, outcomes will be ranked in order of priority by NTREAP (i.e., from 1 (most important) to X (least important)) in the first instance, and then rated according to the GRADE framework (i.e., critical; important, but not critical; of limited importance) by NTWC. The final list of priority populations/conditions, outcome domains and a list of outcome measurement tools (including time points) will be prioritised by NTWC, with feedback from NTREAP on outcome domains considered in final decision making. This list will be derived from the outcomes reported in eligible studies and, where available, core outcome sets for that condition (identified by

searching COMET[5] and ICHOM[6]). To ensure the most relevant outcomes are prioritised, for each condition, core outcomes sets and/ or outcomes from relevant Cochrane reviews (where available) will be visible to the NTREAP and NTWC when conducting the outcome prioritisation process.

Throughout the prioritisation exercise, the NTREAP and NTWC will remain blinded about the characteristics or results of included studies to prevent knowledge of study results or other characteristics (such as study design) from influencing decision-making. In determining the critical and important outcomes, the NTWC will be guided by GRADE (7), and focus on the relevance and validity of outcome measures. Validity and relevance of outcome measures will be determined using outcome measurement instrument hierarchies, where available, from Cochrane groups, core outcome sets and utilising the clinical expertise of NTWC members..

Outcomes reported at different timepoints (across studies) will be grouped and considered as follows: short term, intermediate term, long-term, or not specified. Deciding whether something is considered short, intermediate or long term for a population will be guided by the published evidence, the NTWC and COMET. To avoid unit-of-analysis issues associated with repeated observations within a study (see Section 3.3.9), data from a single time point will be selected for each outcome, as determined by the NTWC during outcome prioritisation (typically immediately post-intervention). Where multiple timepoints are assessed as critical or important to decision making (e.g. short- and long- term remission in symptoms) separate outcomes will be specified for each timepoint.

[5] See *http://www.comet-initiative.org/*

[6] See *https://www.ichom.org/*

3.2 Search methods for identification of studies

3.2.1 Electronic searches

The literature search strategy (see Appendix B) was developed in Ovid (for Embase and MEDLINE) based on the key element of research question (i.e., the intervention). The search is not limited by population or outcome, but rather by study type; with methodological filters for identifying RCTs and exclusions for publication types developed and published previously (66, 67).

In developing the search strategy, we appraised and adapted keywords and MeSH terms previously reported; with the search strategies of SRs identified in the scoping report also reviewed to identify additional potentially relevant concepts (68, 69). Terms or concepts proven not suitable were removed and other terms added.

No date, language or geographic limitations will be applied when conducting the search of English language databases. Non-English databases will not be searched.

The strategy will be adapted to suit the required syntax for the following electronic bibliographic databases:

- Embase (via Ovid)
- MEDLINE (via Ovid)
- Cochrane Central Register of Controlled Trials (via Cochrane Library)
- Emcare (via Ovid) – coverage of all nursing specialty areas
- PsycINFO (via Ovid) – coverage of behavioural science and mental health
- AMED (via Ovid) – coverage of Allied and Complementary Medicine
- CINAHL (via *EBSCOHost*) – Cumulative Index to Nursing and Allied Health Literature
- PubMed (limited to in-process citations and citations not indexed in MEDLINE) – to retrieve citations not yet indexed in OVID
- Pan American Health Organization (PAHO) Virtual Health Library (VHL) – including Lilacs (Health information from Latin America and the Caribbean countries), PAHO IRIS (institutional repository for information sharing), and BRISA (Regional Base of Health Technology Assessment Reports of the Americas)

As the populations for eligible NRSIs will be specified after initial screening of RCTs for prioritised populations with no RCTs (see Section 3.1.2 and Flowchart 2) the search strategy will be implemented in two phases:

1. All eligible RCTs will be found using the search strategy outlined in Appendix B. Independently, populations will be prioritised by the NTWC and NTREAP as outlined in Section 3.1.2 and Flowchart 2.
2. If there are no RCTs identified for a priority population, an additional search for NRSIs in that population will be conducted. This second search will use the search strategy listed in Appendix B, augmented with population-specific search strings. The additional population filters will be developed and tested using the process described above and approved by the NTWC prior to implementation.

3.2.2 Other sources

The public will be invited by the Department of Health to submit references for published research evidence, however any grey literature will be excluded. Reference lists of included studies will not be examined (i.e. backward citation searching) as empirical studies assessing the value of this indicate that it is most useful when reviewing areas that are difficult to search electronically (e.g. new technologies) or for which authors aim to identify grey literature, which is not eligible for inclusion in this review (70).

3.3 Data collection and analysis

As outlined below, included studies will be critically appraised, appropriate data extracted into data extraction tables, and the results analysed and summarised into appropriate categories according to prioritised populations/conditions and comparators. Summary of findings tables will be developed for up to seven critical and important outcomes, guided by the GRADE framework.

3.3.1 Inclusion decisions

Studies identified in the literature searches

Title/abstract screening

Citations (title/abstracts) retrieved by the literature searches will be imported into EndNote and duplicates removed. Citations will then be imported to Covidence (www.covidence.org), an online tool that streamlines the screening and data extraction stages of a systematic review. Initial piloting of the screening process will occur with the first 100 records to ensure consistency. If needed, any screening guidance will be updated prior to screening the remaining citations. Each citation (title/abstract) will be screened by one of three evidence reviewers who will discard ineligible studies (marked as irrelevant and tagged with a reason for exclusion) and retain those with eligible data or information (marked as relevant or maybe). All citations marked as irrelevant will then be screened by a second reviewer to ensure eligibility criteria have been appropriately applied. Where there is uncertainty about relevance, a decision will be made through discussion with the third reviewer, who will decide to either mark the citation as irrelevant or take it through to full text.

Citations that are in a language other than English will be tagged and managed as described below under *Studies published in languages other than English*.

Full text screening

Full text articles identified for possible inclusion in the evidence synthesis will be retrieved and independently assessed for inclusion by two reviewers. A pre-specified, hierarchical approach as outlined in Appendix C will be used to annotate reasons for exclusion, with the results of the study selection process illustrated in a PRISMA diagram. Ineligible studies will be listed in a table in the technical report under *'Characteristics of excluded studies'*, which will include the reason for exclusion. Where there is uncertainty or conflicts about inclusion, a decision will be made through discussion with a third reviewer. If additional expertise or advice about the application of the PICO criteria is required, further follow up with the NTWC will occur (noting that the NTWC will be presented with excerpts from the publication relevant to the query whilst remaining blinded to other identifying details such as the study citation, design, size, risk of bias, or results).

If a study does not contain the required PICO information for a decision to be made about its eligibility, the article will be tagged as 'relevant but additional follow up needed' and the information will be sought from the study's authors, through an open-ended request. These studies may be listed as *'Studies awaiting classification'* or *'Ongoing'* and may be considered when developing conclusions about the *'Overall completeness and applicability of evidence'*.

Trial registration numbers, author names, and study titles, locations and dates will be used to find multiple reports arising from the same study (i.e., protocols, trial registries etc.). These citations will be linked within Covidence, with each study being allocated a unique Study ID and cited in the final report. Published errata or corrigenda identified in the search will be checked and linked to the appropriate study.

Eligible studies that are not available in English will be noted and managed as described in the below under *Studies published in languages other than English*.

Citations referring to clinical trial records will be eligible for inclusion if they meet the *Criteria for considering studies for this review*. If the record is confirmed as meeting the eligibility criteria but published results are not available, it will be included as part of the *'Ongoing'* studies. If a clinical trial record is associated with a published study already identified in this review, the citations will be linked in Covidence as described above.

All eligible studies will be cross checked with the *Retraction Watch* database via *Zotero*.

Evidence provided through the Department's public call for evidence

Potentially relevant primary studies identified by the NTWC, NTREAP, and other key stakeholders will be considered for inclusion if they satisfy the eligibility criteria described above (see *Criteria for considering studies for this review*).

All submitted literature will be collated, tabulated, and cross referenced with the evidence identified in the literature search described in Section 3.3.1 above. In-scope studies not identified in the literature search will be incorporated into the evidence evaluation. A rationale for exclusion (as noted in Appendix C) will be provided for all studies considered out of scope (documented in a table within the technical report).

Studies published in languages other than English

Studies published in languages other than English will undergo title and abstract translation using Google translate. Translated titles and abstracts will be screened to remove irrelevant citations, with articles excluded at title and abstract screen reported in the *'Results of the search'*. If online translation does not facilitate understanding of the title and abstract, then these studies will be listed in a table as *'Studies unable to be translated or interpreted at the title/abstract stage'*.

Translated titles and abstracts will be reviewed and evaluated against the *Criteria for considering studies for this review* as described above. Full text translation will not occur to determine eligibility. Studies assessed as potentially eligible for inclusion in the review will be recorded in a *'Studies Awaiting Classification'* table. This information will also be reflected in the PRISMA flow diagram.

The potential risk of language bias and its implications for the evidence evaluation will be discussed in relevant sections of the Evaluation Report (such as *'Overall completeness and applicability of evidence'* and *'Agreements and disagreements with other studies or reviews'*).

3.3.2 Data collection process

Data extraction will occur in three phases:

1. Preliminary data extraction of Population, Intervention and Comparator criteria of eligible studies. Here, studies will be coded to ICD-11 categories, individualised or non-individualised homeopathy, and either placebo/sham, control (no intervention, wait list, usual care) or other (active comparator). This will be used to inform the population prioritisation process, and address questions relating to appropriate population grouping for the evidence synthesis, including identifying any 'gold standard' active comparators.
2. Data extraction of the study details (priority populations), starting with coding of the outcome domains, measures and timepoints to inform the outcome prioritisation phase, followed by data extraction of the study characteristics of included studies.

3. Data extraction of results for critical (or important) outcomes selected for inclusion in the evidence synthesis and summary of findings tables.

Pre-testing of standard data extraction and coding forms (see Appendix E) will involve all reviewers, who will data extract the required information from the same three primary studies. The studies will be selected to cover different homeopathic medicinal products, conditions and outcomes. The project lead (MJ) will inspect the forms to ensure the relevant data are extracted as intended and any necessary revisions will be made to ensure consistency.

For each included primary study, one reviewer will extract data using the pre-tested data extraction and coding forms. All data extraction forms will be checked for completeness and accuracy by a second reviewer. Where there is uncertainty or disagreement about included data, a decision will be made through discussion with a third reviewer.

3.3.3 Requests for data

Eligible primary studies not published in English, ongoing trials and studies published as conference abstracts with incomplete results will be identified for inclusion. These studies will be noted as '*Ongoing*' or within the '*Studies Awaiting Classification'* table and will not be included in the evidence appraisal. Study authors will not be contacted to obtain further information.

No attempts will be made to obtain or clarify data from published peer-reviewed studies.

3.3.4 Data items

The following characteristics of included studies will be extracted (see Appendix E): study references, study design, year conducted, setting and location, participant characteristics (including demographics, comorbidities, etc.), intervention and comparator characteristics (including name, essential elements of the intervention, materials/procedures, who provided, modes of delivery, where, when, how much, tailoring, modifications, adherence), co-interventions, outcomes (including measurement method, timing), and funding sources.

If a study used (and reported) different approaches to assess the effect of the intervention, we will extract the data based on the following order of preference (71):

1. Full intention-to-treat analysis (i.e., an analysis of participants in the intervention groups to which they were randomised at baseline, regardless of the intervention they received).
   a. If outcome data are missing, imputations for the missing data were made using either:
      i. a model-based approach (e.g., likelihood-based analysis, inverse-probability weighting) (preferred), or
      ii. calculated as if they were observed (e.g., last observation carried forward, mean imputation, regression imputation, stochastic imputation).
2. Modified intention-to-treat analysis (i.e., an analysis that adheres to intention-to-treat principles except certain data are justifiably not included). This includes participants with missing outcome data (no adjustments made), certain patients who never start treatment, and individuals deemed ineligible after randomisation.
3. An 'as-treated' or 'per-protocol' analysis (i.e., an analysis of the effect of adhering to the intervention as described in the trial protocol). This includes participants analysed according to the intervention they received, even if randomised to a different treatment group; or the exclusion of individuals who did not adhere to the assigned intervention.

This approach was selected based on the RoB 2 signalling question, which asks whether an appropriate estimate of effect was used. In the context of this review, the preferred estimate of effect is an intention-to-treat analysis or modified intention-to-treat analysis (71).

3.3.5 Missing data

For studies where the results are incompletely reported (e.g. no effect estimate is reported, but the direction of effect is reported along with a *p*-value), we will report the available information. If the reported information allows for calculation of effect estimates or missing statistics (e.g. SD), we will perform the calculations as described in Chapter 6 of the Cochrane Handbook (78).

Studies with missing outcome data will be included alongside other studies for that condition; either in the narrative (non-quantitative) synthesis of results or on forest plots showing the sample size. Investigations into missing outcomes or results within a study will be conducted through inspection of the clinical trial record or protocol (where available) and will be noted when assessing the risk of bias for that study (see Section 3.3.6).

Missing results from studies awaiting classification (including studies published in a language other than English) or ongoing trials will be considered when interpreting the evidence for each outcome (see Section 3.3.13).

3.3.6 Tools to assess risk of bias in individual studies

The risk of bias of included studies will be assessed using the most appropriate risk of bias assessment tool according to the type of study as follows:

- RCTs: Revised Cochrane Risk of Bias (RoB) tool v2.0 (71, 72)
- NRSIs: ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions (73, 74)

Randomised controlled trials

RoB v2.0 consists of five domains that assess bias arising from the randomisation process: bias due to deviations from intended interventions; bias due to missing outcome data; bias in measurement of the outcome; and bias in selection of the reported result. Each domain will be assessed for bias, which will be recorded as 'high', 'low', or 'some concerns'. Concerns of bias will be raised when it is considered plausible (i.e., likely, probable, possible or conceivable) that bias was present, with the algorithm provided for the RoB v2.0 used to guide decision making (available online at *https://www.riskofbias.info*). Versions of the RoB v2.0 relevant to different study designs (i.e. cluster randomised control trials and crossover trials) will be used where appropriate.

In being consistent with the order of preference for analysis of intervention studies to inform health policy decisions (see Section 3.3.4), as recommended by the Australian Government (75, 76) (when claiming superiority), the Cochrane Handbook (72) and GRADE (7), the risk of bias for domain 2 will be judged according to the effect of assignment to the intervention (the intention-to-treat effect). In this context, it is noted that, there is a potential for bias associated with non-blinding of trial participants or trial personnel (for individualised medicines). However, the only deviations from the intended intervention that will be assessed are:

4. those considered to arise because of the trial context (i.e., unconscious or conscious processes associated with recruitment and engagement activities),

5. those considered to be inconsistent with the trial protocol, and
6. those judged likely to influence the outcome (as per guidance for RoB v2.0) (72).

This means that any deviations considered to occur outside the trial context do not lead to a judgement of bias for the effect of assignment to the intervention (e.g., dropouts due to a change in circumstance that prevents the participants' ability to participate). No hard rule will be set for an expected dropout rate to be considered reasonable, as the impact of missing data may depend on the type of outcome (e.g. continuous or dichotomous) and risk of the event (72).

An overall risk of bias for each outcome in the RCT will be judged based on the following criteria:

- *overall low risk of bias* – low risk of bias for all domains
- *some concerns* – at least one domain has some concerns raised, but none are found to be at high risk of bias
- *overall high risk of bias* – high risk of bias for one or more domains

Nonrandomised studies interventional studies (NRSIs)

Critical appraisal of NRSIs will be guided by the methods described by Cochrane using the ROBINS-I tool (73, 74). Potential confounders and co-interventions for each population identified for inclusion will be identified and agreed through discussion with the NTWC prior to assessment of the risk of bias. ROBINS-I evaluates the risk of bias observed in the following domains: confounding, selection of participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes and selective reporting. Each domain will be judged for risk of bias, which will be recorded as 'low', 'moderate', 'serious', 'critical', or 'no information provided'.

The overall risk of bias judgement for a specific outcome will use the following guide:

- *overall low risk of bias* – the study is comparable to a well-performed RCT and is judged to be a low risk of bias for ALL domains
- *overall moderate risk of bias* – the study appears to provide sound evidence for a nonrandomised study but cannot be considered comparable to a well-performed randomised trial. The study is judged to be a low or moderate risk of bias for ALL domains
- *overall serious risk of bias* – the study has some important problems and is judged to be at serious risk of bias in at least ONE domain, but not a critical risk of bias in any domain
- *overall critical risk of bias* – the study is too problematic with regards to this domain to provide any useful evidence on the effectiveness of the intervention. The study is judged to be at critical risk of bias in at least ONE domain
- *no information* – there is no information on which to base a judgement about overall risk of bias. There is no clear indication that the study is at serious or critical risk of bias AND there is a lack of information in one or more key domains of bias.

Studies rated as at critical risk of bias in any domain will be excluded from the reporting of results, synthesis, and conclusion; however, study details will be included under '*Characteristics of included studies*'.

3.3.7 Risk of bias assessment process

The risk of bias for each included study will be assessed by one reviewer. A second reviewer will then check and confirm all assessments made. Disagreements will be resolved by discussion, with advice sought from a third reviewer if agreement cannot be reached.

The initial assessment will be based on the primary outcome for that study (if a primary outcome is not stated, the assessment will be on the main/key outcome of the report). The second reviewer will check the risk of bias assessment when conducting the evidence synthesis (i.e. when examining the outcome results of the study for inclusion in a meta-analysis and when developing GRADE summary of findings tables), with the focus of the assessment being on the outcome of interest. That is, the second reviewer will check that the 'study level' assessment was appropriate for the outcome, with any additional notes added to the risk of bias comments.

To ensure consistency among reviewers, pre-testing of risk of bias assessments will involve all reviewers, who will complete assessments for the same three RCTs and three NRSIs (using RoBv2.0 and ROBINS-I, respectively). The studies will be selected to cover different homeopathic medicinal products and conditions. The project lead will inspect the forms to ensure consistency, and any differences will be resolved through discussion.

For each outcome we will report our judgement of risk of bias (e.g., low, moderate, high, critical) by domain and provide a rationale for the judgement with supporting information (see Appendix D). Overall risk of bias judgements will be described in the '*Characteristics of included studies*' table, which will be based on the primary (or main) outcome measure for that study.

3.3.8 Measures of effect

Dichotomous data will be presented as risk ratios (RR). Continuous data will be reported as mean difference (MD) (along with the standard deviation (SD) and number of participants). Standardised mean difference (SMD) will be used when different scales are used to measure the same conceptual outcome (e.g. function). To ensure that all the scales point in the same direction of effect, where required, data will be adjusted before standardisation by multiplying the mean value by -1 to be consistent with other studies. Time-to-event data will be presented as hazard ratios (HR) and, if analyses of covariance have been used to adjust for baseline measures, the adjusted effect estimates will also be recorded. Count data will be presented as a rate ratio. For all measures of effects, we will report 95% confidence intervals and *p*-values.

To reduce effects of confounding, adjusted effect estimates from NRSIs (if included) will be reported, if available (e.g. adjusted odds ratios (OR) from logistic regression or adjusted rate ratios from Poisson regression analyses). The variables that have been used for adjustment will be recorded. If multiple adjusted effect estimates are reported, the most appropriate result to be reported will be made in collaboration with the NTWC.

As there are a broad range of populations eligible for inclusion in the review, it is not possible to pre-specify the minimal clinically important differences (MCID) or specific thresholds for interpreting effect size for each outcome. However, where possible, the MCID will be sourced from published reports or will be guided by advice from the NTWC, taking care to note if the reported value is a minimal detectable change (MDC: statistical) (i.e., the smallest change in score that likely reflects true change more than measurement error alone); or the minimal clinically important difference (MCID: clinical) (i.e., the smallest difference between the scores in a questionnaire that the patient

perceives to be beneficial). For each outcome, details of the measure and whether the MCID or MDC is known will be recorded. Where the MCID/MDC is available, the magnitude of effect (if important) will be considered on three levels: small (MD <10% of the scale), moderate (MD between 10% to 20% of the scale), or large (MD more than 20% of the scale). If an important effect is not reached, then the effect will be considered trivial ("little or no").

In the absence of an MCID or when SMD is used for different measures of an outcome, Cohen's guidance for interpreting the magnitude of the SMD will be used: 0.2 represents a small effect, 0.5 a moderate effect, and 0.8 a large effect (77). For dichotomous outcomes, a 25% relative risk reduction or increase will be used as a threshold.

3.3.9 Unit of analysis issues

No imputation for unit of analysis issues will be performed.

While intervention-related clustering is a possibility in this review, it is considered unlikely that included studies will have provided adequate details to enable clusters to be accounted for in their analyses. No adjustments will be made for intervention-related clustering using a statistical method. However, where such clusters are identified, this will be noted in the relevant part of the review along with discussion of the potential impacts of the clustering on the review findings.

Cluster-randomized trials

To avoid a unit-of-analysis error in a cluster-randomised trial we will report effect estimates from analyses undertaken by the trial authors. Information regarding the approach used to account for the cluster design will be recorded. If the study authors have not provided information relating to the method of adjustment (i.e., the estimate of the relative variability within and between clusters), the implications of the missing data will be considered when interpreting the evidence and will be discussed under '*Overall completeness and applicability of evidence*'.

Crossover trials

To avoid a unit-of-analysis error in a crossover trial, only data from the first period (i.e., before crossover has occurred) will be included in the analysis. Studies reporting paired analysis will be discussed separately, and the potential impact of selective reporting will be discussed under '*Overall completeness and applicability of evidence*'.

Repeated observations

To avoid a unit-of-analysis error in studies reporting data for more than one timepoint (i.e., repeated measures during treatment and at end-of-treatment), results from a single timepoint will be selected, and only data from that timepoint will be presented in the analysis. The timepoint selected will be based on that determined to be critical (or important) for decision making as outlined in Section 3.1.4.

3.3.10 Studies with more than two intervention groups

If the included studies have multiple treatment groups, only single pairwise comparisons of the intervention with a control (i.e. 'no intervention' or 'other') will be considered. If appropriate to combine like groups (e.g., a homeopathic medicinal product at different potencies or dosing frequency), we will combine to create a single pair-wise comparison. The combining of summary statistics across groups will be as described in Chapter 6 of the Cochrane Handbook (78).

3.3.11 Quantitative synthesis

Given the size and breadth of this review, it is expected that a broad approach to data synthesis will transpire. That is, summary estimates will include an outcome domain (e.g., pain) measured at a rough time point (e.g., latest timepoint reported), using any instrument (e.g., visual analogue scale or numeric rating scale). This approach is intended to capture as many studies as possible for any given PICO.

Synthesis (meta-analysis) will be undertaken for studies that compare homeopathic medicinal products with 'placebo' or 'control'. Results data from studies comparing homeopathic medicinal products with other 'active' interventions will not be extracted, except if requested by the NTWC.

The NTWC may request that data comparing a homeopathic medicinal product with another active intervention be synthesised for a prioritised population/condition, where:

- at least two studies compare the effect of homeopathic medicinal products with the same active comparator, and the comparator is sufficiently homogenous across studies to support synthesis, and
- at least two of these studies are at low or moderate risk of bias, and
- the comparator represents an accepted, evidence-based 'gold standard' of care for the population in question.

Such cases will be identified by the NTWC through blinded discussions with the evidence reviewer, with a list of active comparators identified from included studies provided to the NTWC prior to the data synthesis stage (see Appendix A – Frameworks).

Data from RCTs

Data synthesis from RCTs only will be performed using RevMan 5.4 (58). Within each comparison (PICO) we will combine effect estimates across studies for each outcome using a random effects model to account for expected differences between studies. Forest plots will be stratified by individualised vs non-individualised (vs not specified, if not all studies include this information). Statistical heterogeneity will be assessed by visually inspecting the overlap of confidence intervals on the forest plots, formally testing for heterogeneity using the Chi^2 test (using a significance level of $\alpha=0.1$), and quantifying heterogeneity using the I^2 statistic (79).

Effect estimates will not be combined across outcomes if analysis of covariance has been used to adjust for baseline measures, or for time-to-event data reported as hazard ratios.

Data from NRSIs (priority populations where no RCTs identified only)

If applicable, for those priority populations for which NRSIs are included, data synthesis from NRSIs will be performed using RevMan 5.4 (where appropriate) and forest plots will be presented. Within each comparison we will combine effect estimates across studies for each outcome using a random effects model to account for expected differences between studies. Forest plots will be stratified by individualised vs non-individualised (vs not-specified, if not all studies include this information) Statistical heterogeneity will be assessed by visually inspecting the overlap of confidence intervals on the forest plots, formally testing for heterogeneity using the Chi^2 test (using a significance level of $\alpha=0.1$), and quantifying heterogeneity using the I^2 statistic (79).

Effect estimates will only be combined across outcomes if the included NRSIs are judged to be at low to moderate risk of bias (see Section 3.3.6) and are sufficiently homogenous to be combined. This

means the PICO criteria of the NRSIs must be sufficiently similar and the study design features should be comparable.

3.3.12 Non-quantitative synthesis

The evidence review will provide a structured narrative summary of the results for each condition identified (including study design and population demographics) along with risk of bias assessments and other intervention characteristics. This will be followed by a summary of results grouped by comparator and outcome domain. Results from each study will be reported, with the range and magnitude of observed effects noted.

Results tables will be structured by comparator ('control' or 'other' intervention), outcome domain, and study design and will be ordered by study ID (author, date). Where possible, a visual representation of the results of included studies will be presented in a forest plot (without a summary estimate) grouped by risk of bias. Any important differences in study design or features that may influence the interpretation of results will be considered and discussed in the text. To describe an overall effect across multiple studies for each outcome (for studies comparing homeopathic medicinal products with control only), we will describe the observed effects across the studies using a simple vote count based on direction of effect (e.g., X/Y studies reported an effect favouring the intervention for the outcome Z).

Qualitative descriptors of the size of the effect (small, large etc.) will be used only in relation to the evidence statement and will be based on the smallest difference that patients perceive as beneficial (or detrimental) for that outcome (i.e., MCID), as described in Section 3.3.8.

3.3.13 Risk of reporting bias across studies

Judgements regarding missing data across studies will be made based on available information (e.g., eligible studies that are listed as *'Ongoing'* or '*awaiting classification'* including studies published in a language other than English) and discussed under '*Overall completeness and applicability of evidence*'. The implications of the missing data will also be considered when interpreting the evidence the GRADE summary of findings (publication bias) (see Section 3.3.16). Here, judgements will be made on whether the missing data is due to non-reporting biases (e.g. the result is not favourable to the intervention or non-significant) and whether any additional data would likely alter the summary estimate. Approaches for assessing bias due to missing studies (such as forward searching clinical trial registers, grey literature, or other reports) will not be performed.

If more than 10 RCTs are included for a particular PICO, funnel plots (of effect estimates against their standard errors) will be generated in RevMan 5.4 to determine possible non-reporting bias ('unknown-unknowns'). If, after visual inspection of the funnel plot there is evidence of asymmetry (suggesting small-study effects or missing results), a brief statement about the potential impact on the overall conclusions of the evidence review will be included under the relevant sections of the review (including the *'Overall completeness and applicability of evidence'*). Other possible reasons for funnel plot asymmetry will also be considered at this time (e.g., fitting fixed-effect model, poor methodological quality, true heterogeneity, chance) (80). No additional statistical analysis for testing for small-study effects will be conducted.

3.3.14 Addressing risk of bias

All RCTs will be included in the review, regardless of judgements made regarding risk of bias. A description of the risk of bias in individual domains will be presented along with the estimated effect estimate. To examine the impact of study bias, a sensitivity analysis will be conducted, with studies judged to be at high risk of bias to be removed from the analysis (see Section 3.3.11). The impact of this change on the effect size will be noted and considered when interpreting the certainty of evidence (risk of bias across studies) (see Section 3.3.16).

NRSIs rated as at being at serious or critical risk of bias across one or more domains will not be included in the reporting of results, synthesis, and conclusion. A brief statement about the impact of the exclusion of NRSIs at serious or critical risk of bias on the overall conclusions of the evidence review will be included under the relevant sections of the review (including the '*Overall completeness and applicability of evidence*').

3.3.15 Subgroup analyses

Subgroup analysis will be used to explore possible sources of statistical heterogeneity relating to the delivery of homeopathy but will not be conducted on subsets of participants (e.g. men vs women etc.). Studies will be grouped according to whether homeopathy was delivered as individualised or non-individualised homeopathic prescription (see Section 1.2 Description of the intervention). A standard test (see Section 3.3.11 Quantitative synthesis) for heterogeneity across subgroups will be reported.

3.3.16 Certainty of the evidence

Across each population, we will assess the certainty of the evidence for each outcome using the GRADE approach (7). Evidence from RCTs and NRSIs (where included) will be evaluated separately as will evidence comparing homeopathy with 'placebo' and 'control'.

The GRADE process provides a framework for determining the certainty of the evidence and is based on consideration of the following five factors:

- *Risk of bias.* Based on the summary assessment across studies for each outcome reported for a comparison (81). Concerns relating to bias will be raised when sensitivity analyses suggest RCTs at high risk of bias are influencing the result (see Section 3.3.14) or when all studies contributing data are at high risk of bias for that outcome.
- *Inconsistency.* Based on heterogeneity in the observed intervention effects across studies that suggests important differences in the effect of the intervention and whether this can be explained (82). Concerns relating to inconsistency will likely be raised if there is unexplained non-overlap of confidence intervals or if statistical tests suggesting substantial heterogeneity cannot be explained (see Section 3.3.11 and Section 3.3.15). Where a result is based on a single study, inconsistency will not be rated.
- *Imprecision.* Based on interpretation of the upper and lower confidence limits in relation to a minimal clinically important threshold (e.g. the confidence interval includes both appreciable benefit and harm); and whether the optimal information size has been reached (i.e. the total number of patients meets the required sample size for a sufficiently powered individual study). In the absence of a published clinically important threshold a rough guide will be used (e.g. the confidence interval overlaps with small or large difference based on Cohen's guidance for interpreting SMD (77), or a 25% relative risk reduction or increase) (83).

- *Indirectness.* Based on important differences between the review questions and the characteristics of included studies that may lead to important differences in the intervention effects (84). Concerns relating to indirectness will likely be raised if delivery of the homeopathic medicinal product is not consistent with practice in the Australian setting.
- *Publication bias.* Based on the extent to which the evidence is available. Publication bias would be suspected when the evidence is limited to a small number of trials reporting a positive result or there are concerns about missing results (85) (see Section 3.3.13).

The certainty of evidence will be categorised as follows:

- High (⊕⊕⊕⊕): further research is very unlikely to change the confidence in the estimate of effect.
- Moderate (⊕⊕⊕⊖): further research is likely to have an important impact in the confidence in the estimate of effect.
- Low (⊕⊕⊖⊖): further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
- Very low (⊕⊖⊖⊖): any estimate of effect is very uncertain.

For each factor, a judgement will be made about whether there are serious, very serious or no concerns – resulting in an overall GRADE describing the certainty of evidence for each outcome. Footnotes will be used to record judgements made about downgrading (or upgrading) the evidence (see Section 3.3.17). Scoring of the certainty of the evidence will begin as 'high' for RCTS and NRSIs (score=4), which can be downgraded by –1 for each domain with serious concerns or –2 for very serious concerns (7, 86).

The certainty of evidence will also be upgraded in certain (exceptional) circumstances as per GRADE guidance (7). Increasing the certainty of evidence would only be made if no serious limitations are noted in any of the factors mentioned above and would generally only apply to NRSIs (7). Factors that will be considered for upgrading the evidence include the following:

- *Large magnitude of an effect.* When large or very large effect estimates are observed, and there is more confidence in the results (enough events to be precise).
- *Dose-response gradient.* When there is a clear relation between outcome and increasing exposure levels
- *Effect of plausible residual confounding.* If there are clear factors that have likely led to an under-estimate of the true effect such as unmeasured or unknown determinants in the adjusted analysis that are likely to be distributed unequally between intervention and control groups (7).

3.3.17 'Summary of findings' table

For each population, findings for the critical and important outcomes (see Section 3.1.4), will be reported in summary of findings tables that will be prepared using the GRADEpro GDT software (*www.gradepro.org*). Estimates of treatment effects for each outcome will be reported as absolute and relative risks (or SMD). When quantitative data is not available, a narrative synthesis will be provided (see Section 3.3.12). All critical and important outcomes will be reported, regardless of whether the findings demonstrate a clinically meaningful change.

The summary of findings tables will provide a summary of each of the included outcomes and the certainty of evidence rating for each outcome in a quick and accessible format (7). As part of the summary of findings table, an evidence statement pertaining to each outcome will be included. This statement will be guided by the prescribed format provided in GRADEpro (87), with the preferred statement to be selected listed in Table 1.

Table 1 List of informative statements to communicate results of systematic reviews

SIZE OF THE EFFECT ESTIMATE	SUGGESTED STATEMENTS *
HIGH Certainty of the evidence	
Large effect	X results in a large reduction/increase in outcome
Moderate effect	X reduces/increases outcome
Small important effect	X reduces/increases outcome slightly
Trivial, small unimportant effect or no effect	X results in little to no difference in outcome
MODERATE Certainty of the evidence	
Large effect	X probably results in a large reduction/increase in outcome
Moderate effect	X probably reduces/increases outcome
Small important effect	X probably results in a slight reduction/increase in outcome
Trivial, small unimportant effect or no effect	X probably results in little to no difference in outcome
LOW Certainty of the evidence	
Large effect	The evidence suggests X results in a large reduction/increase in outcome
Moderate effect	The evidence suggests X results in a reduction/increase in outcome
Small important effect	The evidence suggests X results in a slight reduction/increase in outcome
Trivial, small unimportant effect or no effect	The evidence suggests that X results in little to no difference in outcome
VERY LOW Certainty of the evidence	
Any effect	The evidence is very uncertain about the effect of X on outcome

Source: modified from Santesso et al. (2020) (87)
* Replace X with intervention, replace 'reduce/increase' with direction of effect, replace 'outcome' with name of outcome, include 'when compared with Y' when needed)

A technical report that presents, in detail, the evidence base for each research question by outcome will be developed and will include the following information:

- The methodology used to identify the evidence base (documented systematic literature search, inclusion and exclusion criteria described).
- The characteristics of included studies, data extraction and risk of bias forms.
- Detailed results, presented by outcome, which will contain detailed information about the evidence assessment.

Contributions of authors

MJ wrote and developed the draft Research Protocol with contributions from MI, AM, AT, and MI in writing sections, providing comment, testing the data extraction and screening process, and proofreading final drafts. The search strategy was developed and tested by MJ and MI.

NTREAP, homeopathic researcher Dr Robbert van Haselen and NTWC provide expert advice, especially in relation to intervention, study design, and eligibility criteria. Cochrane Australia conducted a methodological review of the draft protocol.

Declarations of interest

This work is paid for by the National Health and Medical Research Council (NHMRC) under contract number 2020-21C009.

All named authors declare they have no financial, personal or professional interests that could be construed to have influenced the development of this systematic review protocol. In line with NHMRC processes, prior to entering a contract, NHMRC considers any interests disclosed by external contractors. Accordingly, prior to entering the contract, NHMRC considered the interests disclosed by HTAnalysts. NHMRC determined that such interests did not preclude the engagement of HTAnalysts.

In addition, in line with the process to establish any NHMRC committee, each committee member was asked to disclose their interests. Declarations of interest among NHMRC Natural Therapies Working Committee members are lodged with the NHMRC and are available *online*.

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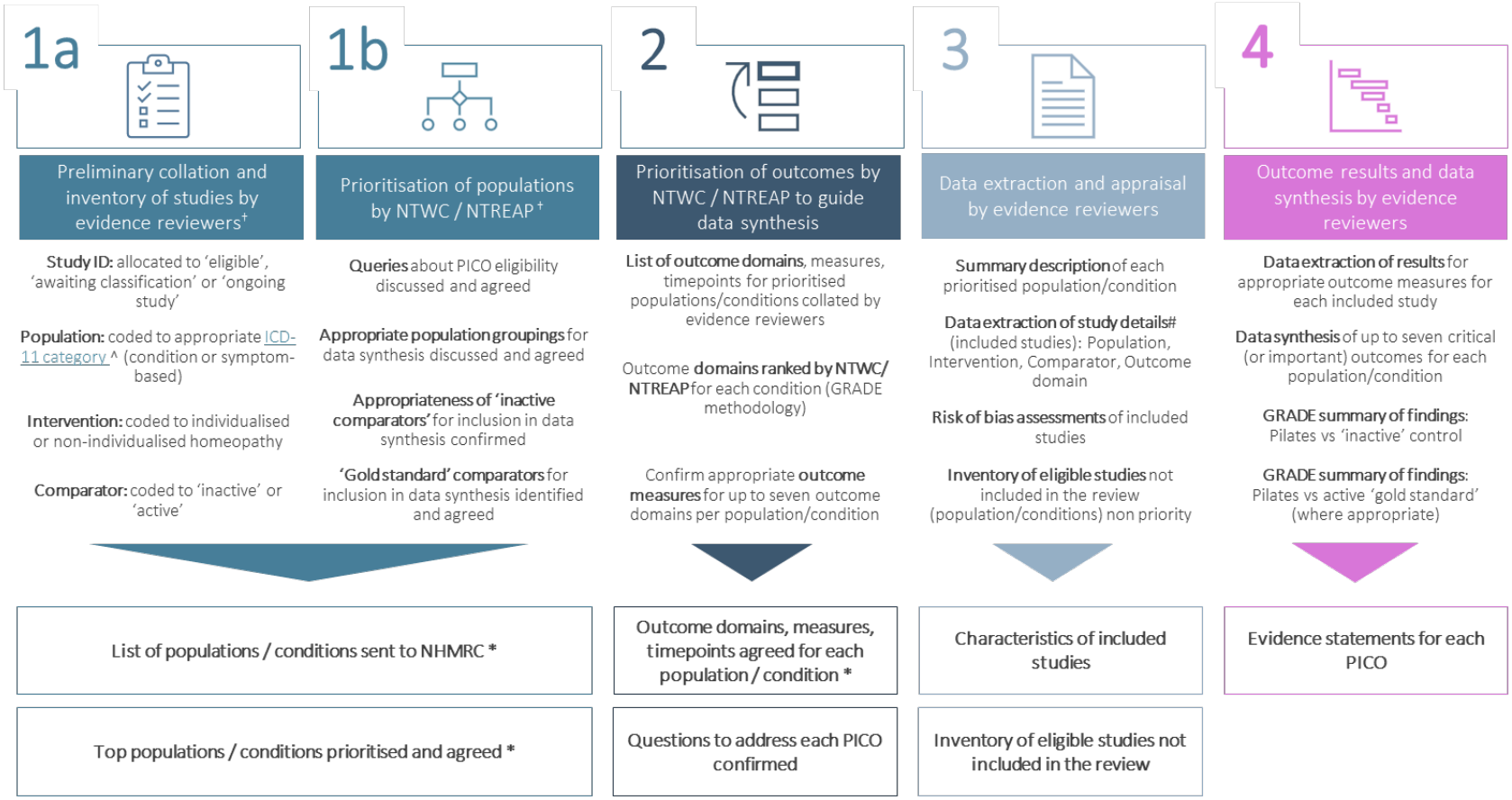
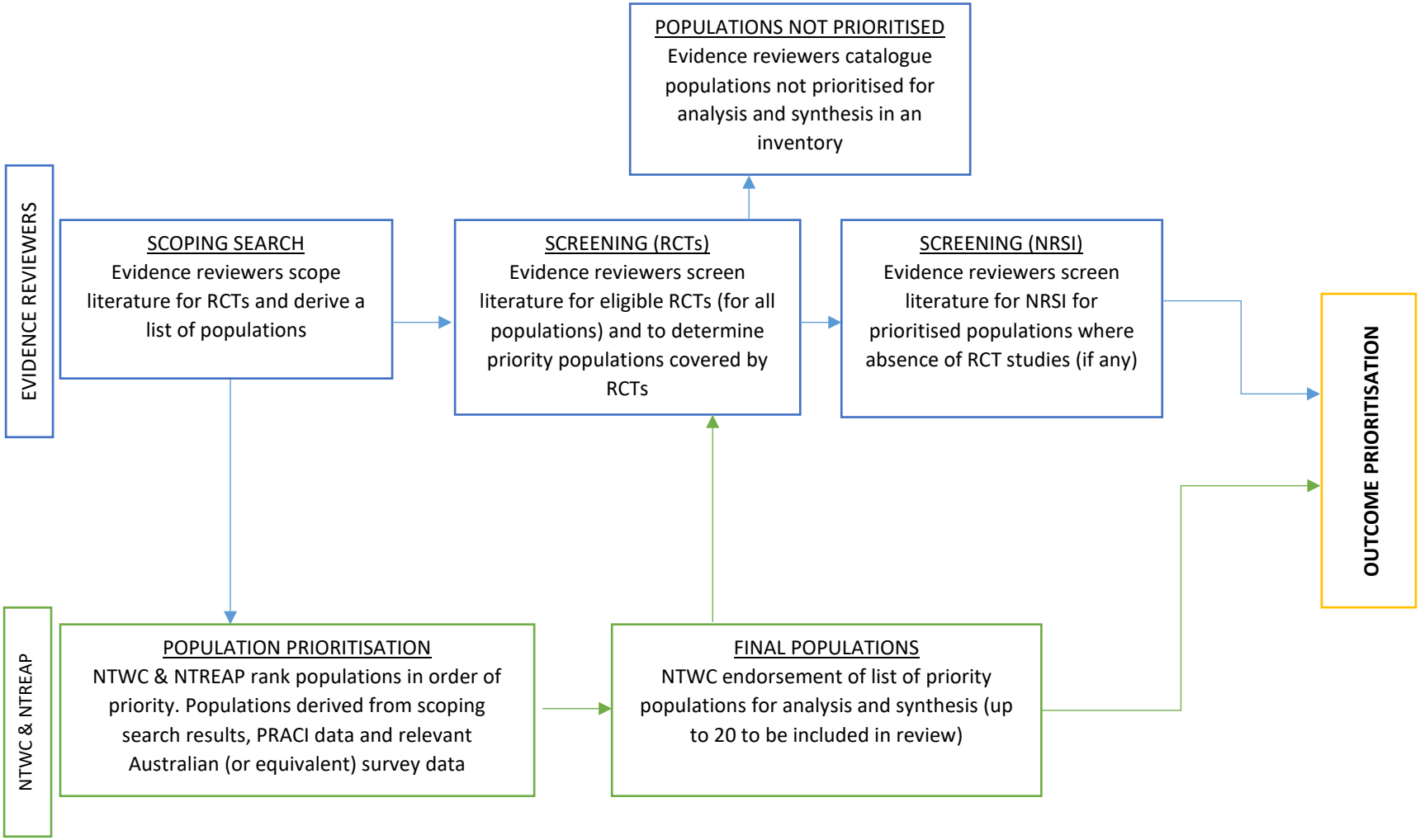
Appendix A – Frameworks

Flowchart 1 **Framework for refining the research question and data synthesis**



Notes:
† Step 1a and 1b to occur simultaneously. NTWC/NTREAP to prioritise populations independently from evidence reviewers collating inventory of RCTs.
^ ICD-11, International Classification of Diseases for Mortality and Morbidity Statistics (ICD-11 MMS) 11th Revision (available at https://icd.who.int/browse11/l-m/en)
* No identifying information about study ID, study design, study size, study quality or outcome results available (see Framework 2).
# Preliminary data extraction of included studies will begin at step 2 to inform outcome domains.

Flowchart 2 Framework for population prioritisation



Appendix B - Literature search strategy

Concept: Study design limits (SRs, RCTs, NRSIs, not case reports, nonhuman etc.)

1. exp meta analysis/ or meta analysis.mp. or exp systematic review/ or systematic review.mp. or pooled analysis.mp. or ((exp review/ or review.mp.) and (systemat* or pool*).mp.)
2. exp comparative study/ or comparative study.mp. or exp clinical trial/ or clinical trial.mp. or randomized controlled trial.mp. or randomi?ed controlled trial.mp. or exp randomized controlled trial/ or exp randomisation/ or randomization.mp. or randomi?ation.mp. or exp single blind procedure/ or single blind procedure.mp. or exp double blind procedure/ or double blind procedure.mp. or exp triple blind procedure/ or triple blind procedure.mp. or exp crossover procedure/ or crossover procedure.mp. or exp placebo/ or placebo*.mp. or random*.mp. or rct.mp. or single blind.mp. or single blinded.mp. or double blind.mp. or double blinded.mp. or treble blind.mp. or triple blind.mp. or triple blinded.mp. or exp prospective study/ or prospective study.mp.
3. case report/
4. (editorial or letter or comment or historical article).pt.
5. (animals/ or nonhuman/) not humans/
6. or/3-5

Concept: Homeopathy

7. homeopathy/
8. homeopathic agent/
9. Materia medica/
10. (materia medica or nosode*).ti,ab,kw.
11. (dilut* adj2 (very or ultra* or high or serial* or substance* or agent*)).ti,ab,kw.
12. (potentis* or potentiz*).ti,ab,kw.
13. (homeopathy or homeopathic or homeopathia or homeopath*).ti,ab,kw.
14. (homoeopathy or homoeopathic or homoeopathia or homoeopath*).ti,ab,kw.
15. or/6-13

Concept: RCTs for screening

16. (15 AND 2)
17. 16 NOT (1 OR 6)

The above strategy is designed for OVID (Embase and Medline) will be adapted to suit EBSCO (CINAHL, AMED), the Cochrane Library and PubMed (limited to in-process citations and citations not indexed in MEDLINE).

Ovid syntax

Exp explodes controlled vocabulary term (i.e. includes all narrower terms in the hierarchy)

* denotes a term that has been searched as a major subject heading

/ denotes controlled vocabulary terms (EMTREE)

$ truncation character (unlimited truncation)

$*n* truncation limited to specified number (*n*) of characters (e.g. time$1 identifies time, timed, timer, times but not timetable)

* truncation character (unlimited truncation)

? substitutes any letter (e.g. oxidi?ed identifies oxidised and oxidized)

adj*n* search terms within a specified number (*n*) of words from each other in any order

.ti. limit to title field

.ti,ab. limit to title and abstract fields

.kw,ti,ab. limit to keyword, title and abstract field

.pt limit to publication type

CINHAL syntax

* truncation character (unlimited truncation)

wildcard character will replace 1 or 0 characters (e.g. f#etus will retrieve fetus and foetus)

? wildcard character will replace one character (e.g. wom?n will retrieve women and woman)

MH - Search the exact CINAHL® subject heading; searches both major and minor headings

MH"heading"+ Search an exploded subheading

TI search title fields

AB search abstract fields

Nn – Proximity "near" operator will find a result if the terms are within a certain number (n) words of each other, regardless of the order in which they appear. (e.g. eating N5 disorders for results that contain eating disorders, as well as mental disorders and eating pathology.)

PT limit to publication type

PubMed syntax

The PubMed search will be restricted to records that are not indexed for MEDLINE (i.e. in-process citations and citations from journals (or parts of journals) that are not currently MEDLINE-indexed)

The search will comprise free-text terms only and replicates the free-text sets in the Embase search (converted from the Ovid syntax).

* truncation character (unlimited truncation)

[TI] limit to title field

[TIAB] limit to title and abstract fields

[EDAT] date citation added to PubMed

[SB] PubMed subset

AND pubmednotmedline[sb] will be added to the last line of search string

Appendix C – Screening criteria

A priori screening criteria are listed below.

Items 1 through 9 will be considered and applied at abstract/title screening. At full text screening, all items will be considered and applied as appropriate (these studies will be listed in the technical report with reasons for exclusion):

1. Duplicate citation
2. Nonhuman study
3. Intervention out of scope (not homeopathy or component thereof)
4. Comparator out of scope (study compares different homeopathic medicinal product 'head-to-head')
5. Population out of scope (healthy participants seeking general wellness, no identifiable risk factors)
6. Outcome out of scope (patient experiences/preferences, safety, quality and economic outcomes)
7. Publication type out of scope (specify)
   a. opinion piece/editorial/commentary
   b. not an intervention study examining effectiveness (e.g. surveys)
   c. grey literature
8. Study design out of scope (specify)
   a. General review, Guideline, HTA assessment
   b. SR of RCTs or NRSIs
   c. Case series or other
9. Other (specify) [b]
   a. withdrawn
   b. erratum
10. Awaiting classification (relevant but additional follow up needed (specify) [c]
    a. Nonrandomised comparative study
    b. conference proceeding (data incomplete)
    c. Publication not available in English [a]
11. Ongoing study (results not available)

a. Articles published in a language other than English will be considered as described in the Section *'Studies published in languages other than English'.*

b. Articles tagged as 'other' will be included and linked to the primary study to enable identification of all associated citations.

c. Articles tagged as relevant but additional follow up needed are included but will not be incorporated in the evidence appraisal. These studies may be listed as '*Studies awaiting classification*', '*Ongoing*', or may be considered when developing conclusions about the '*Overall completeness and applicability of evidence*'.

Appendix D – Risk of bias forms

Cochrane RoB v2.0 (Randomised controlled trials)

Study ID		
Domain	Judgement	Summary description [a]
Bias arising from the randomisation process	High risk Some concerns Low risk	Describe the method used to determine if: • the allocation sequence was random; • the allocation sequence was adequately concealed; • baseline differences between intervention groups suggest a problem with the randomisation process.
Bias due to deviations from intended interventions	High risk Some concerns Low risk	Describe the method used to conceal treatment allocation: • were participants aware of their assigned intervention during the trial? • Were carers and people delivering the interventions aware of participants' assigned intervention during the trial? • Consider the potential effect of deviations due to assignment and deviations due to adherence
Bias due to missing outcome data	High risk Some concerns Low risk	Describe the completeness of outcome data for each primary/secondary outcome, including whether: • data for this outcome were available for all, or nearly all, participants randomised; • (if applicable) there was evidence that the result was not biased by missing outcome data; • (if applicable) missingness in the outcome was likely to depend on its true value (e.g. the proportions of missing outcome data, or reasons for missing outcome data, differ between intervention groups).
Bias in measurement of the outcome	High risk Some concerns Low risk	Describe whether: • the method of measuring the outcome was inappropriate; • measurement or ascertainment of the outcome could have differed between intervention groups; • outcome assessors were aware of the intervention received by study participants; • (if applicable) assessment of the outcome was likely to have been influenced by knowledge of intervention received.
Bias in selection of the reported result	High risk Some concerns Low risk	Describe whether: • the trial was analysed in accordance with a pre-specified plan that was finalised before unblinded outcome data were available for analysis; • the numerical result being assessed is likely to have been selected, based on the results, from multiple outcome measurements within the outcome domain; • the numerical result being assessed is likely to have been selected, based on the results, from multiple analyses of the data.
Overall risk of bias		

Source: Chapter 8 Cochrane Handbook for Systematic Reviews of Interventions (72).

Notes:

a. See www.riskofbias.info for the precise wording of signalling questions and guidance for assessing risk of bias. The most up to date RoB 2 tool specific for the study design (e.g., cross-over) will be used.

ROBINS-I (nonrandomised studies of interventions)

Study ID		
Domain	Judgement	Comments
Bias due to confounding	Low risk Moderate risk Serious risk Critical risk No information	Describe whether: • there is potential for confounding of the effect of intervention in this study; • the analysis is based on splitting participants' follow up time according to intervention received; • intervention discontinuations or switches were likely to be related to factors that are prognostic for the outcome; • the authors use an appropriate analysis method that controlled for all the important confounding domains; • the variables adjusted for are valid and reliable measures of the confounding domains; • the authors control for any post-intervention variables that could have been affected by the intervention; • the authors use an appropriate analysis method that adjusted for all the important confounding domains and for time-varying confounding; • confounding domains that were adjusted for measured validly and reliably by the variables available in this study;
Bias in selection of participants into the study	Low risk Moderate risk Serious risk Critical risk No information	Describe whether: • the selection of participants into the study (or into the analysis) based on participant characteristics observed after the start of intervention; • the start of follow-up and start of intervention coincide for most participants; • adjustment techniques used are likely to correct for the presence of selection biases;
Bias in classification of interventions	Low risk Moderate risk Serious risk Critical risk No information	Describe whether: • the intervention groups are clearly defined; • the information used to define intervention groups is recorded at the start of the intervention; • classification of the intervention status has been affected by knowledge of the outcome or risk of the outcome;
Bias due to deviations from intended interventions	Low risk Moderate risk Serious risk Critical risk No information	Describe whether: • deviations from the intended intervention is beyond what would be expected in usual practice; • deviations from intended intervention is unbalanced between groups and likely to have affected the outcome; • important co-interventions balanced are across intervention groups; • intervention is implemented successfully for most participants; • study participants adhere to the assigned intervention regimen; • an appropriate analysis used to estimate the effect of starting and adhering to the intervention;
Bias due to missing data	Low risk Moderate risk Serious risk Critical risk No information	Describe whether: • outcome data is available for all, or nearly all, participants; • participants are excluded due to missing data on intervention status; • the proportion of participants and reasons for missing data are similar across interventions; • results were robust to the presence of missing data.
Bias in measurement of outcomes	Low risk Moderate risk Serious risk	Describe whether: • the outcome measure has been influenced by knowledge of the intervention received; • outcome assessors are aware of the intervention received by study participants;

Study ID		
Domain	Judgement	Comments
	Critical risk No information	• the methods of outcome assessment are comparable across intervention groups; • any systematic errors in measurement of the outcome related to intervention received.
Bias in selection of the reported result	Low risk Moderate risk Serious risk Critical risk No information	Describe whether the reported effect estimate is likely to be selected, based on the results: • multiple outcome measurements within the outcome domain; • multiple analyses of the intervention-outcome relationship and different subgroups.
Overall risk of bias		

Abbreviations:
Source: Adapted from Sterne 2019 *(74)*
Notes:
a. See www.riskofbias.info for the precise wording of signalling questions and guidance for assessing risk of bias. The most up to date ROBINS tool specific for the study design features will be used.

Appendix E – Data extraction forms (examples)

Characteristics of included studies (example)

Study ID	Author date					
Study reference	Include all citations, protocol, registry details					
Study design	RCT/NRSI			List any notable features: e.g., crossover, quasi-randomised		
Author affiliation	List all authors affiliations					
Source of funds	List source of funding (or not specified)					
Declared interests of study authors	List declared interests (or none declared)					
Setting / provider (single centre, multicentre, country/s)	List setting (e.g., university hospital, nursing home, community)			List provider details (e.g. Single provider of intervention)		
Country(s) / region	List country (e.g., Tuscany region, Italy)					
Enrolment period	List enrolment dates (e.g., Aug 2014 to Mar 2016)					
Length of treatment and follow up	List treatment duration (weeks or months) and if any follow-up (e.g., 6 weeks with follow-up at 6 and 12 months)					
Description of population	*N=*	*Description*				
# participants	53	Provide brief details (e.g., Elderly patients with osteoporosis at-risk of falls)				
details	Inclusion criteria: Exclusion criteria:					
Description of intervention/ comparator *(as per TIDieR checklist)*	*n=*	*Description (include brief name, elements essential to the intervention, materials/procedures, who provided, modes of delivery, where, when, how much, tailoring, modifications, adherence)*				
Intervention						
Comparator #1 (control)						
Comparator #2 (other)						
add rows as needed						
Co-interventions						
Was the intervention individualised homeopathy?	Yes/ No/ Not specified		Provide details of practitioner (i.e. expertise, background and any specific training)			
Does the study assess homeopathy delivered for primary prevention?	Yes/ No / Uncertain (seek advice)		Provide details			
Is comparator clearly inactive? (select from list)	Yes/ No / Uncertain (seek advice)		Provide details			
Outcomes	*Description (outcome measure, timing, measurement tool, details, other)*					
#1 (example information is provided)	Primary / Secondary/ Not specified	Pain intensity	Baseline, end of treatment (12 wks), follow-up (24 wks)	Numerical pain rating scale (0-100)	higher score means worse pain	e.g., follow-up data missing for 23 participants

Study ID	Author date					
#2						
#3						
#4						
#5						
(add rows as needed)						
Method of analysis						
Statistics	Provide details (e.g., Descriptive, student t-test, ANOVA)					
Population analysed	ITT / PP / Other	Provide details (e.g., Modified intent-to-treat, 2 participants without final assessment data were excluded from the final analysis). Included details of reported similarity/difference between methods (if available)				
Missing data	Provide details (e.g., imputations, loss to follow up, last observation carried forward) Included details of reported similarity/difference between methods (if available)					

Abbreviations: ANOVA, one-way analysis of variance; ITT, intent-to-treat; PP, per protocol
Notes:

Characteristics of studies awaiting classification

Study ID	Author date
Study description	Title or other descriptive text
Study design	RCT
Participants	
Intervention	
Comparator	
Outcomes	
Notes	e.g., Article written in Korean with no English abstract - awaiting translation

Outcome data from included studies* (example)

Study ID	Population	Comparison	Outcome	Timing	Measured with	Measure details	# participants (N)	[intervention] n/N (%) or mean (SD)	[comparator] n/N (%) or mean (SD)	Point estimate (95% CI)	*p*-value	Direction of effect	RoB#
Smith 2007	Anxiety (adolescents 12-18 yrs.)	Homeopathy vs control	Anxiety	baseline, end-of-treatment (8wks)	GHQ-28	mean change from baseline	50	−3.5 (0.34)	−0.32 (0.18)	MD not reported	0.001	Favours intervention	High
			Depression	baseline, end-of-treatment (8wks)	GHQ-28	mean change from baseline	50	−3.2 (0.35)	−1.32 (1.2)	MD not reported	0.003	Favours intervention	High
			[outcome #3]	NR						RR		Not reported	Low
			[outcome #4]	NR						OR		Favours comparator	Some concerns
			[outcome #5]										
			[outcome #6]										
			[outcome #7]										

*Data as reported by study authors; # as assessed in this review (see Appendix C)
Abbreviations: CI, confidence interval; GHQ-28, General health questionnaire; MD, difference between the means NR, not reported; OR, odds ratio; RoB, risk of bias; RR, relative risk; VAS, visual analogue scale; yr, year;
Notes:
a. Mean change from baseline calculated as the mean scores of QOL dimensions at the end of treatment minus the mean scores of QOL dimensions at the beginning of the study.